

P8109 Spring 2026 Homework #1

Due on February 5 at 11:59pm

1. Let $Z_i, i = 1, \dots, 6$, be iid $N(0, 1)$ random variables. Using a statistical software, calculate

$$P\left(\sum_{i=1}^6 Z_i^2 < 6\right), \quad P\left(\sum_{i=1}^4 Z_i^2 > 3 \sum_{i=5}^6 Z_i^2\right).$$

2. Let X be a random variable having an F-distribution with ν_1 numerator and ν_2 denominator degrees of freedom. Show that $Y = 1/X$ has an F-distribution ν_2 numerator and ν_1 denominator degrees of freedom.

3. Let $T = Z/\sqrt{W/\nu}$, where $Z \sim N(0, 1)$ and W is an independent χ_ν^2 -distribution.

(a) Write down the distribution of T .

(b) Using the result that

$$E(W^\alpha) = \frac{2^\alpha \Gamma(\frac{\nu}{2} + \alpha)}{\Gamma(\frac{\nu}{2})}, \quad \nu > -2\alpha,$$

show that

i. $ET = 0$ for $\nu > 1$.

ii. $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$.

4. The problem provides a proof of the Central Limit Theorem for the Poisson distribution, i.e., if $X \sim \text{Poisson}(\lambda)$, then as $\lambda \rightarrow \infty$, $(X - \lambda)/\sqrt{\lambda} \rightarrow_d N(0, 1)$.

(a) Define $X^* = (X - \lambda)/\sqrt{\lambda}$. Show that

$$Ee^{tX^*} = e^{-t\sqrt{\lambda} - \lambda + \lambda e^{t/\sqrt{\lambda}}}.$$

(b) Show that as $\lambda \rightarrow \infty$,

$$Ee^{tX^*} \rightarrow e^{t^2/2}.$$

5. Prove from first principles that, if $X \sim \text{binomial}(n, p)$, then

$$\frac{X - np}{\sqrt{np(1-p)}} \rightarrow_d N(0, 1).$$

Hint: You may want to use the fact that X can be written in distribution as the sum of n i.i.d. Bernoulli random variables, i.e., $X =_d Y_1 + \dots + Y_n$, where $P(Y_i = 1) = p, P(Y_i = 0) = 1 - p$.

6. Let $Y_1, \dots, Y_5$ be a random sample of size 5 from $N(0, 1)$, and let $\bar{Y}_5$ be its sample mean. Let Y_6 be another independent observation from the same population. Identify with justification of the following distributions.

- (a) $\bar{Y}_5$.
- (b) $U = \sum_{i=1}^5 Y_i^2$.
- (c) $V = \sum_{i=1}^5 (Y_i - \bar{Y}_5)^2$.
- (d) $W = \sum_{i=1}^5 (Y_i - \bar{Y}_5)^2 + Y_6^2$.

7. The coefficient of variation (CV) of a distribution F is defined as

$$CV = \frac{\sigma}{\mu},$$

where μ and σ^2 are the mean and the variance of F , respectively. For a random sample $Y_1, \dots, Y_n$ from F , define the sample coefficient of variation as

$$\widehat{CV} = \frac{S}{\bar{Y}},$$

where S is the sample standard deviation and $\bar{Y}$ is the sample mean. Suppose $Y_i \sim N(\mu, \sigma^2)$, $i = 1, \dots, n$.

- (a) Find the distribution of $\widehat{CV}$.
 - (b) Using a statistical table to find the number c such that $P(|\widehat{CV}| \leq c) = 0.95$.
8. Let $X_1, \dots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ random variables. Define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

- (a) Identify the distribution of S^2 .
- (b) Show that

$$\sqrt{n} \left(\frac{S^2}{\sigma^2} - 1 \right) \rightarrow_d N(0, 2).$$

9. Verify whether each of the following distributions belongs to the exponential family:

- (a) Binomial(n, p);
- (b) Gamma(α, β).

10. (a) Find a function $h(X)$ such that $h(\mu) + \frac{1}{2}h''(\mu)\text{Var}(X)$ is not a good approximation of $E[h(X)]$.

- (b) Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ , variance σ^2 , and finite fourth moment. Suppose h is a real-valued function whose first four derivatives $h'(x)$, $h''(x)$, $h'''(x)$, and $h^{(4)}(x)$ exist for all x . Furthermore, suppose that

$$|h^{(4)}(x)| \leq M \quad \text{for all } x,$$

for some $M < \infty$. Then,

$$E[h(\bar{X})] = h(\mu) + \frac{\sigma^2}{2n} h''(\mu) + R_n,$$

where $R_n = O(1/n^2)$.